

MRPL Mangalore Refinery and Petrochemicals Limited

Major Accident Hazard (MAH) Factory Annual Safety Audit Report · 2026

Audit Date: 14-February-2026 | Document Ref: MRPL-MAH-AUDIT-2026-REV3

Audited Facilities: Crude Distillation (CDU-1/2), FCCU, Delayed Coker (DCU), LPG Tank Farm

1. Executive Summary & Regulatory Authority

This annual statutory safety audit was conducted in compliance with Rule 68-J of the Karnataka Factories Rules, 19

2. General Audit Observations

The refinery demonstrated robust process safety management practices across primary crude units. However, spec

FINDING 01: Pressure transmitter PT-101 on FCCU Air Blower discharge exhibited heavy atmospheric salt corrosion on flange bolts.

Corrective Action: Replace carbon steel studs with Inconel 625 bolting before next quarterly inspection. Target: 30 days.

FINDING 02: Emergency Shutdown Valve XV-301 hydraulic actuator recorded a closing stroke time of 5.8 seconds during annual proof test.

Non-Conformance: Exceeds mandatory SIL-3 threshold of 3.0 seconds maximum. Solenoid valve dump port partially restricted by particulate residue.

Immediate Action: Flush hydraulic fluid, replace 24 VDC dual solenoid pilot assembly, and re-test within 48 hours.

FINDING 03: Hydrocarbon gas detector GS-402 in LPG pump house showed zero drift exceeding 15% LEL calibration limit.

Corrective Action: Sensor recalibrated with 50% LEL methane test gas; replaced sintered flame arrestor element.

FINDING 04: Furnace F-101 pass flow meter FT-204 transmitter terminal block showed moisture ingress due to defective cable gland.

Corrective Action: Resealed cable entry with flameproof compound per OISD-156 refinery electrical guidelines.

Audited By: Chief Inspector of Factories & Lead Process Safety Auditor

Signed & Sealed: Verified Field Copy ■ Certified Confidential